

L3: Pointclouds

- Unstructured collection of points
 - x,y,z coordinates in some coordinate system
 - optional: additional parameters, such as color (r,g,b)

Scanners

- Panorama laser scanners
- Mobile mapping system
- UAV Lidar System
- Phenotype Field Robot

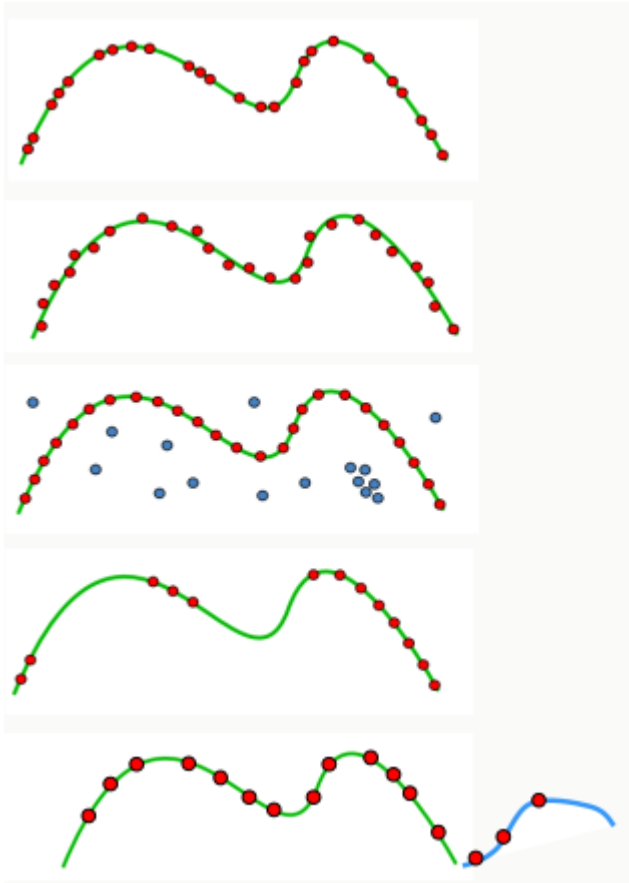
Objectives

- 3D Model of House
 - Civil engineering: BIM - Building information models
 - Digital twins (3D+information)
- Simulation & Gaming
- Industry: Factory /Inventory
- Information about geological formations
 - Geo & Earth sciences
 - Shape & behaviour
 - Land slide monitoring
- Change Detection
 - Construction
 - Mining
- Measure status of crops in the field
 - Growth curves
 - Leaf angle distribution

Properties of Point Clouds

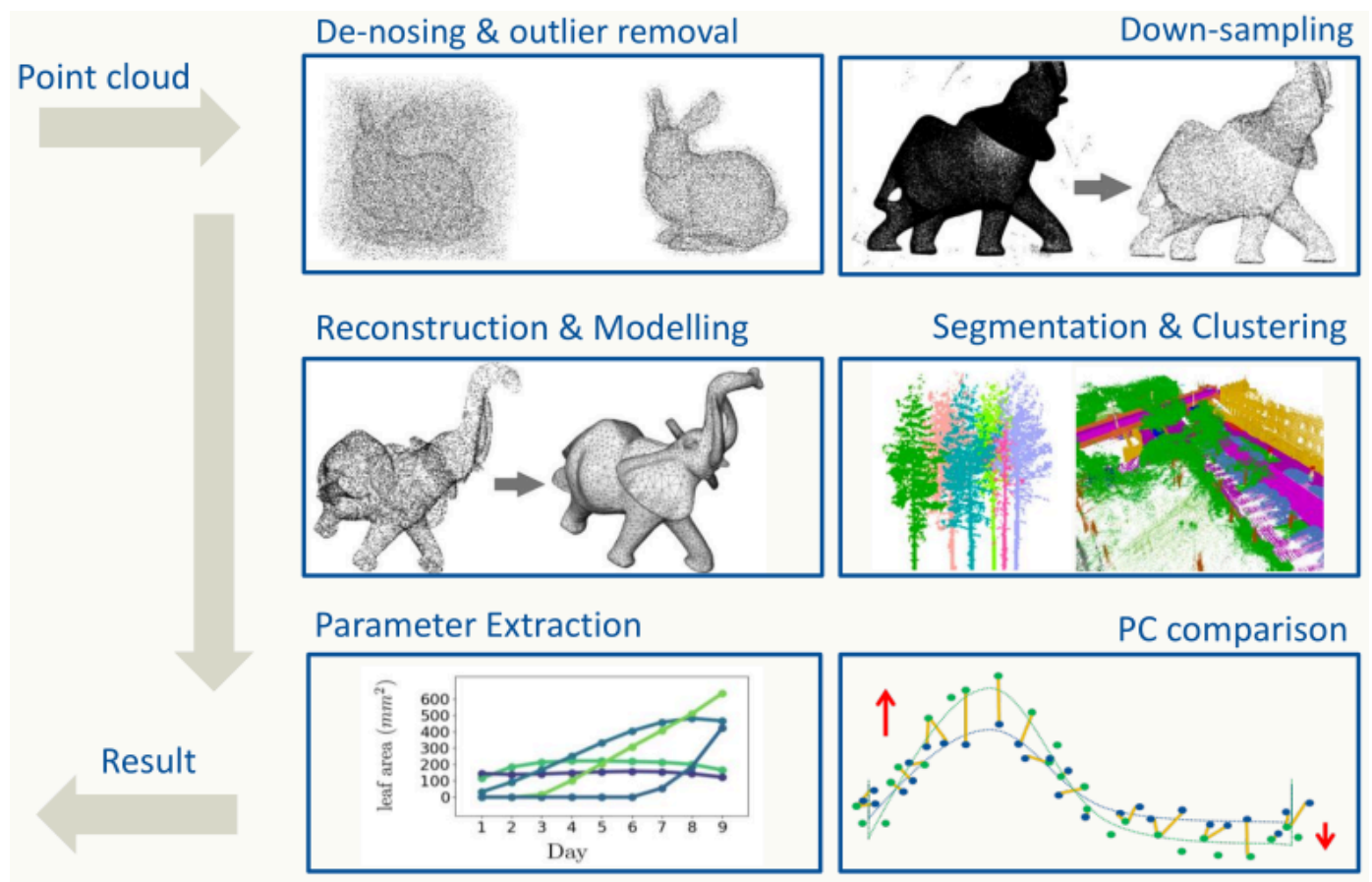
1. Varying spatial point density
2. Noisy points, no repetition measurements
3. High ratio of outliers
4. Many data gaps, occlusions

5. Many objects **not** of interest



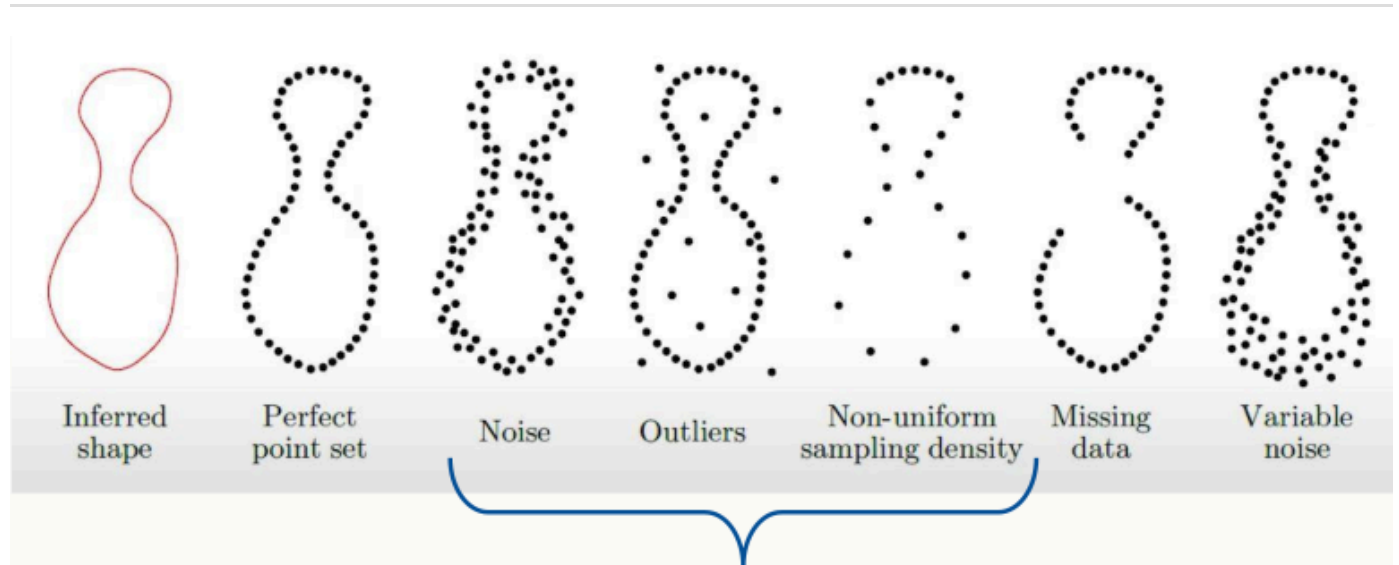
- -> Point-cloud processing needs to account for

PC Processing



- Data structures
 - Octree
 - k-d tree (3D)
 - Efficient low level processing
 - neighborhood/range searching

Preprocessing

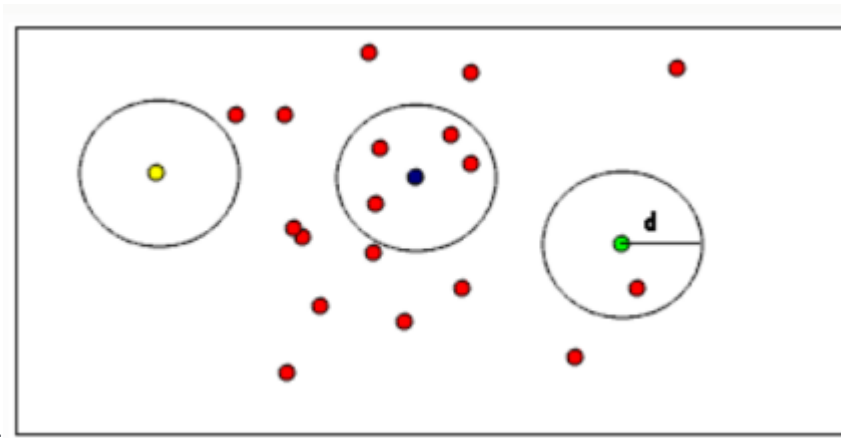


- Outlier removal
- Smoothing/denoising
- Downsampling

Outlier Removal

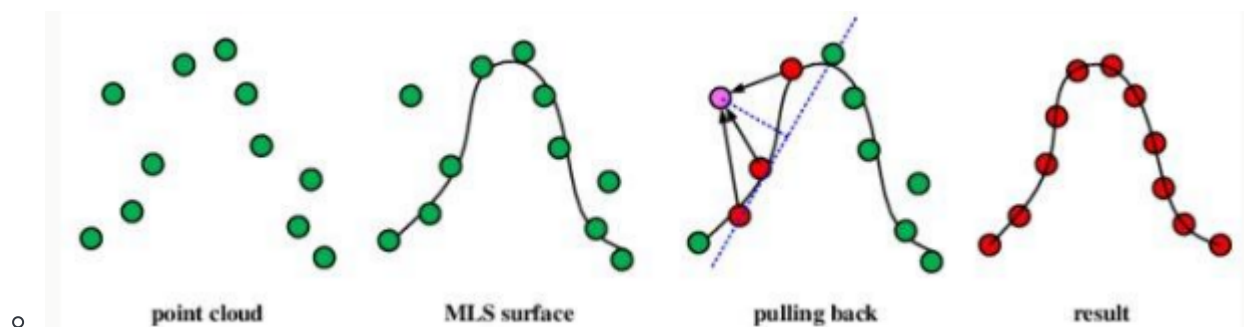
- Ratio of outliers large (many possible causes)
- If concentrated locally: Hard to detect & impact further modeling
- Aim: Point cloud free from outliers
- Elimination by:
 - Geometry: Neighborhood
 - distance
 - density
 - Spectral data
 - intensity
 - RGB
 - Manually
- Example: **Statistical Outlier Removal**

- Remove Points with $< n$ neighbors in sphere with radius r



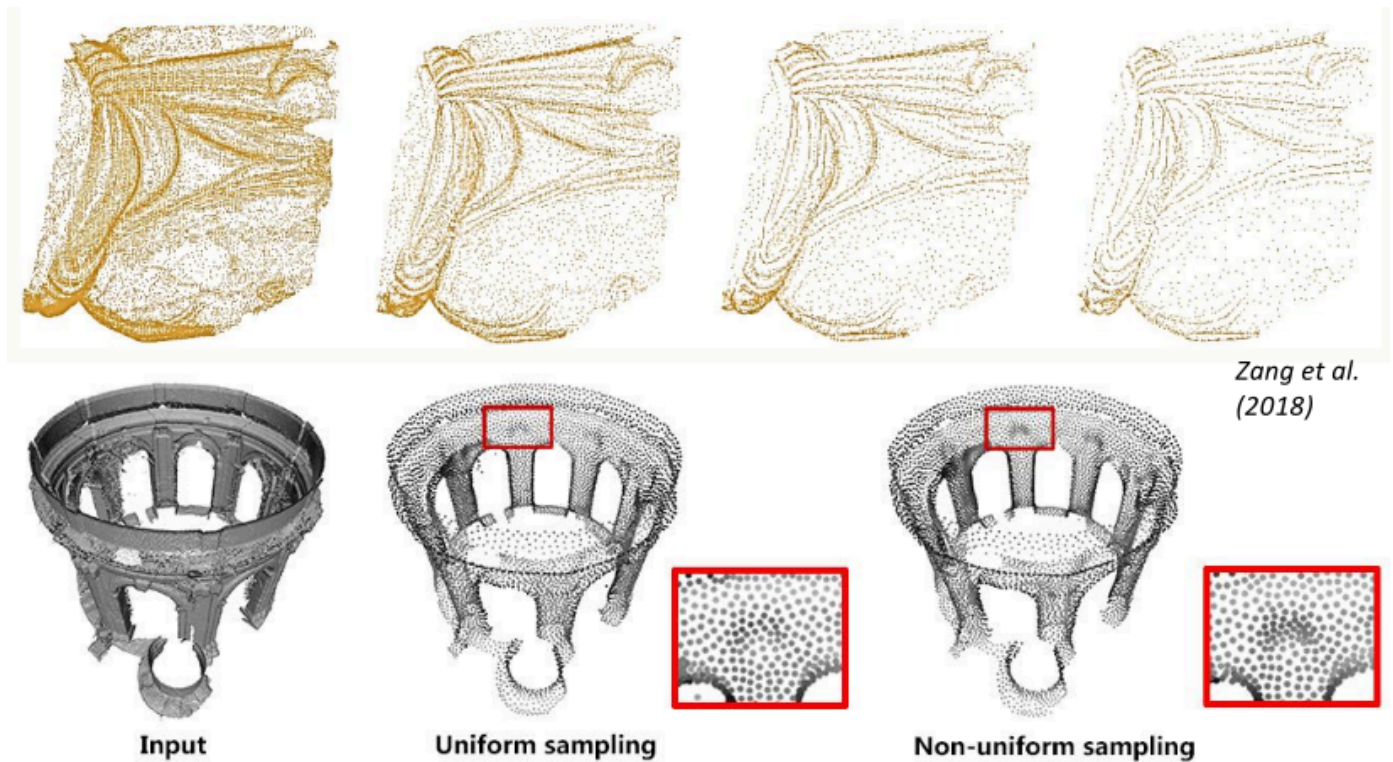
De-Noising/Smoothing

- De-noising difficult
- Requires either:
 - Simultaneous surface reconstruction
 - Simultaneous sub-sampling
 - A priori knowledge of noise
- Noise is not necessarily constant



Down-Sampling

- Data reduction, sub-sampling or down sampling
- From tenths/hundreds of GB to manageable data size
- From irregular to regular distribution
- Procedure: Elimination/selection of points
- Down-sampling+de-noising
 - Local averaging of points
- Strategies:
 - random
 - uniform
 - curvature-based



Segmentation and Clustering

- Segmentation into useful pieces
- Removal of unwanted objects in scene (e.g. background)
- Different modelling/reconstruction procedures for different objects in scene
- Identify stable/static or unstable/moving regions/objects

Machine Learning

- Supervised (learn from input-output pairs)
 - Model from **labeled data**
- Unsupervised (observing degree of similarity, e.g. clustering)
 - Model from **unlabeled data**
- Semi-supervised, Reinforcement learning
- Classical (design meaningful Features for training)
- Deep Learning (let model figure out the features)

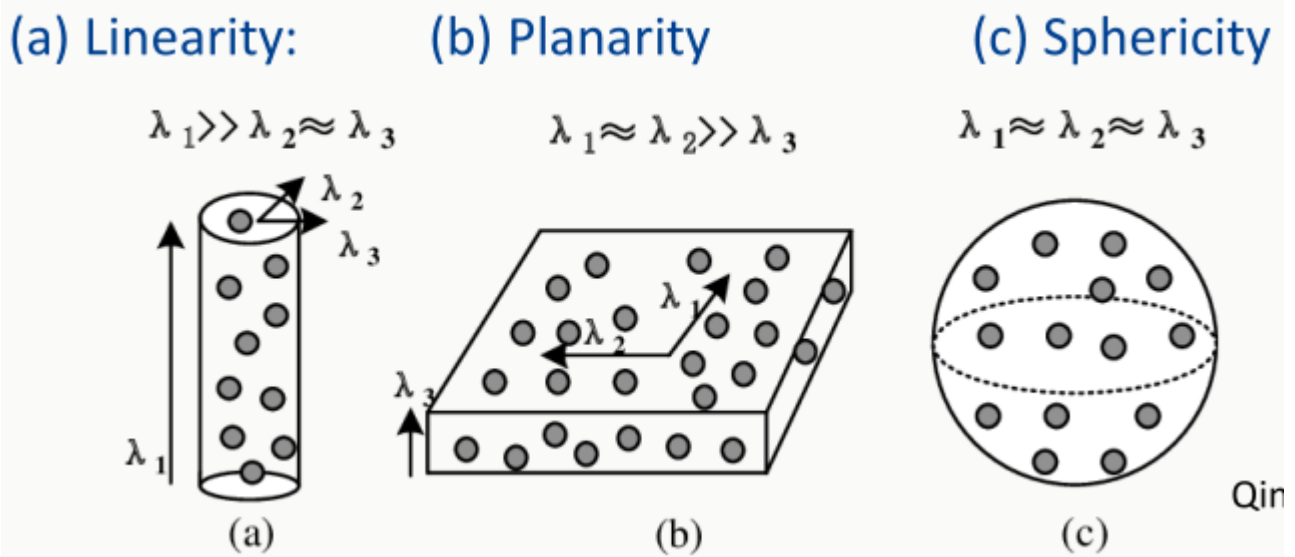
ML Examples

- Supervised
 - Support Vector Machines
 - Decision Trees /Random Forest
- Unsupervised
 - Clustering

- kmeans
- dbscan

Features Examples

- Features describe points and properties
 - Based on neighborhood around point
 - e.g.: within radius r or nearest k points
- Spectral features based on distribution of neighbourhood points
 - Covariance $\rightarrow$ Eigenvalues λ_i



• Linearity:

• Planarity:

• Sphericity:

• Omni-variance:

• Anisotropy:

• Eigenentropy:

• Curvature:

$$f_{L,i}(\mathbf{X}_i, \mathcal{N}_i) = \frac{\lambda_{1,i} - \lambda_{2,i}}{\lambda_{1,i}}$$

$$f_{P,i}(\mathbf{X}_i, \mathcal{N}_i) = \frac{\lambda_{2,i} - \lambda_{3,i}}{\lambda_{1,i}}$$

$$f_{S,i}(\mathbf{X}_i, \mathcal{N}_i) = \frac{\lambda_{3,i}}{\lambda_{1,i}}$$

$$f_{O,i}(\mathbf{X}_i, \mathcal{N}_i) = \sqrt[3]{\lambda_{1,i} \lambda_{2,i} \lambda_{3,i}}$$

$$f_{A,i}(\mathbf{X}_i, \mathcal{N}_i) = \frac{\lambda_{1,i} - \lambda_{3,i}}{\lambda_{1,i}}$$

$$f_{E,i}(\mathbf{X}_i, \mathcal{N}_i) = - \sum_{j=1}^3 \lambda_{j,i} \ln(\lambda_{j,i})$$

$$f_{C,i}(\mathbf{X}_i, \mathcal{N}_i) = \frac{\lambda_{3,i}}{\lambda_{1,i} + \lambda_{2,i} + \lambda_{3,i}}$$

- Other 3D features:

- SIFT

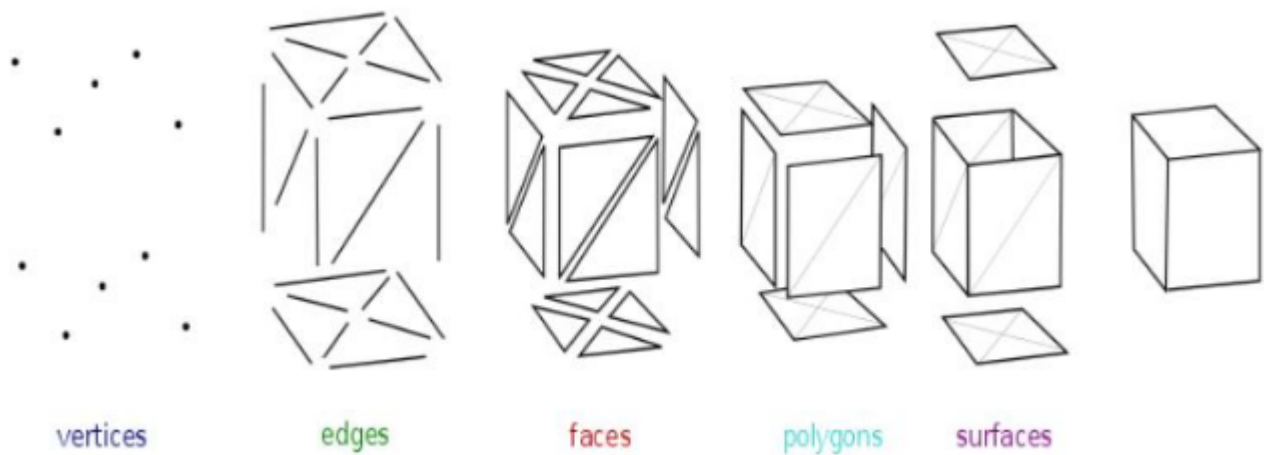
- RIFT
- Fast point feature histogram PFH & FPFH
- VFH: Viewpoint feature histogram
- **Point Structure** for Deep Learning Example:
 - Voxel
 - Projected Images
 - Specific point cloud structure
 - No structure -> original points

Surface Reconstruction

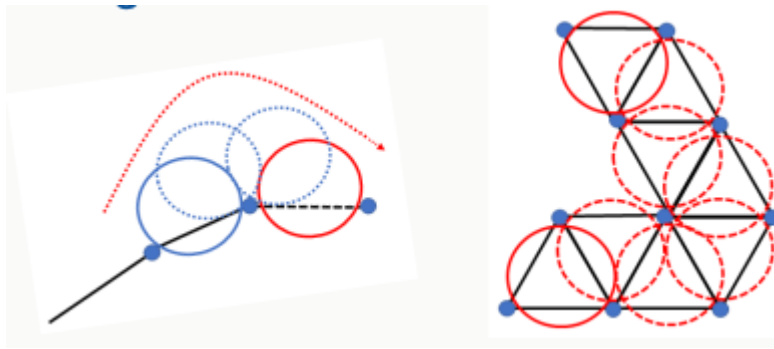
- Find representation of information on surfaces/objects suitable for solving the actual task
- Reduce amount of data
- Integrate previous knowledge about the surfaces /objects
- Fill gaps, smooth, interpolate

Meshes

- Collection of vertices, edges and faces
 - define polyhedral object shape
- Faces: triangles, quadrilaterals, other simple convex polygons



- -> non-parametric surface representation
 - data-driven
 - make no fixed assumptions about population
- **Algorithms:**
 - Ball pivoting
 - Rolling ball with given radius over point cloud
 - Ball touches 3 points -> Create triangle



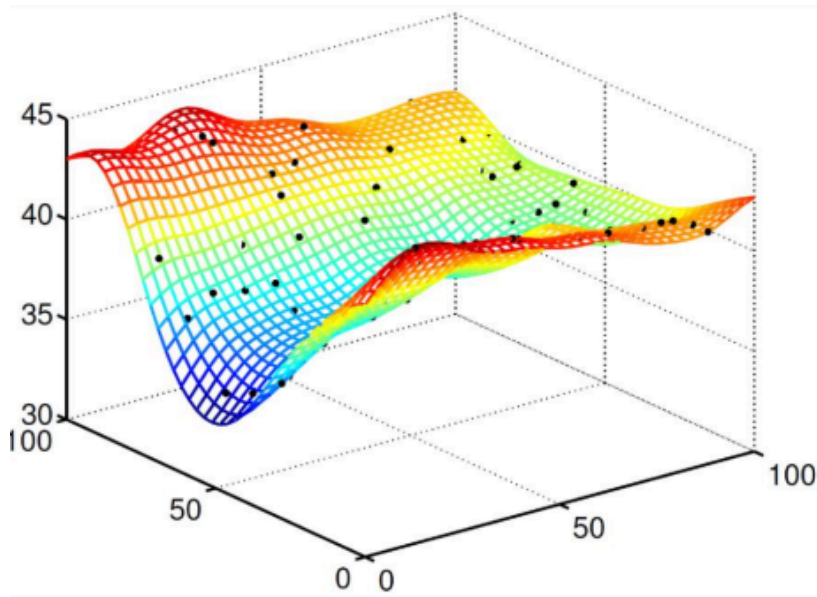
- Marching cubes
- Alpha shapes
- Poisson Surface Reconstruction
- Delaunay 2.5D
- Greedy surface triangulation
- ...

Geometric Primitives

- Spheres:
 - Parametric
 - Regular geometrical shapes
 - Strong prior about the object shape
 - $[x_M, y_M, z_M, r]$
- Planes: $[n_x, n_y, n_z]^T$
- Cylinders: $[x_0, y_0, r, \phi_x \phi_y]$
- Rotational Paraboloid - like Antennae

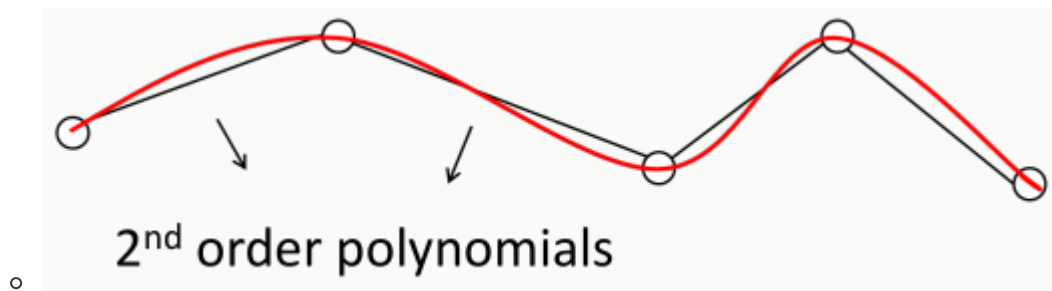
Surface Reconstruction: Freedom Surfaces

- Fitting Multi-dimensional polygons to model the surface
- e.g. 3rd order $z = a_1 + a_2x + a_3x^2 + a_5xy + a_6y^2 + a_7x^3 + a_8x^2y + a_9xy^2 + a_{10}y^3$

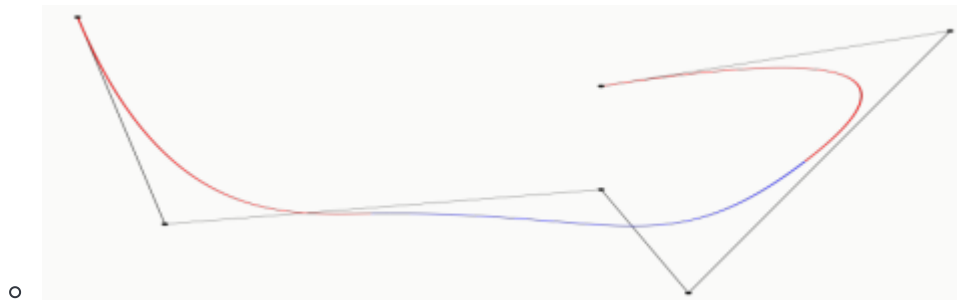


„parametric“ surface representation

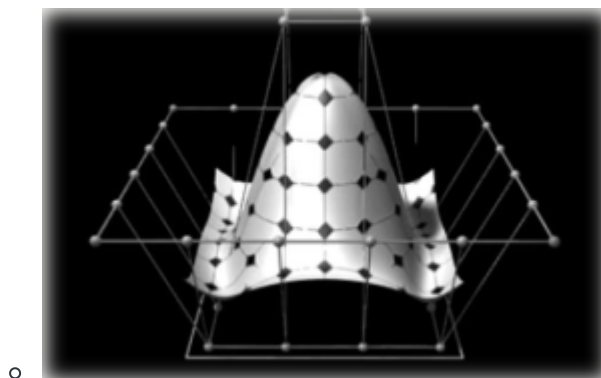
- Splines



- B-Splines (Basis Splines)



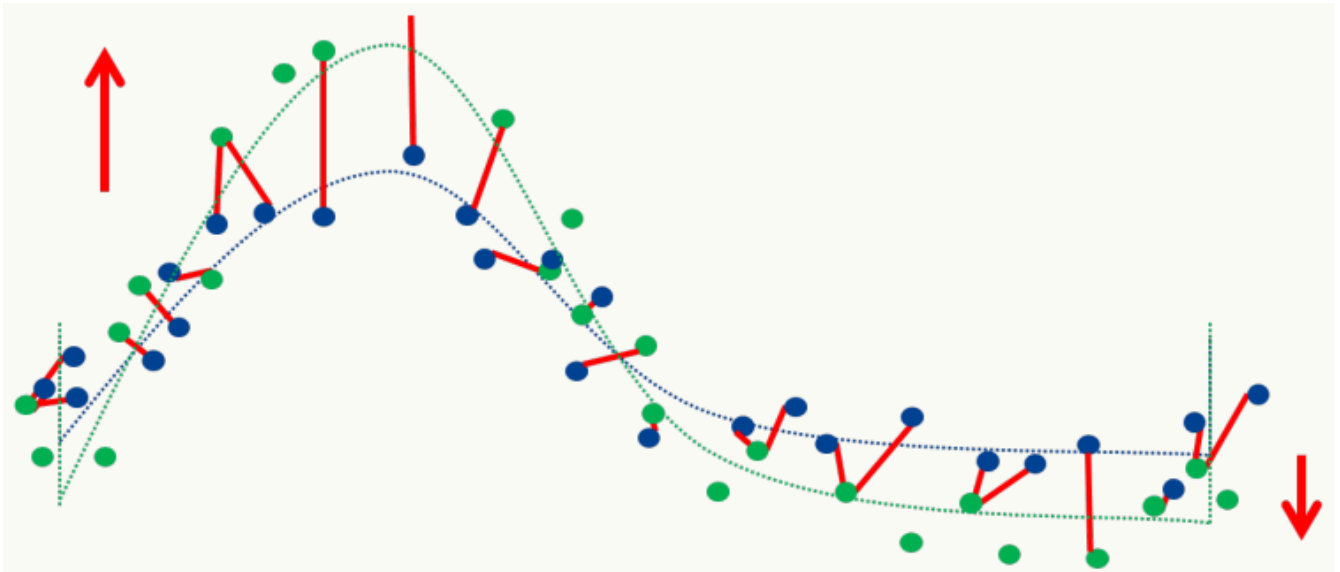
- Non-uniform rational B-splines (NURBS)



Comparing Point Clouds

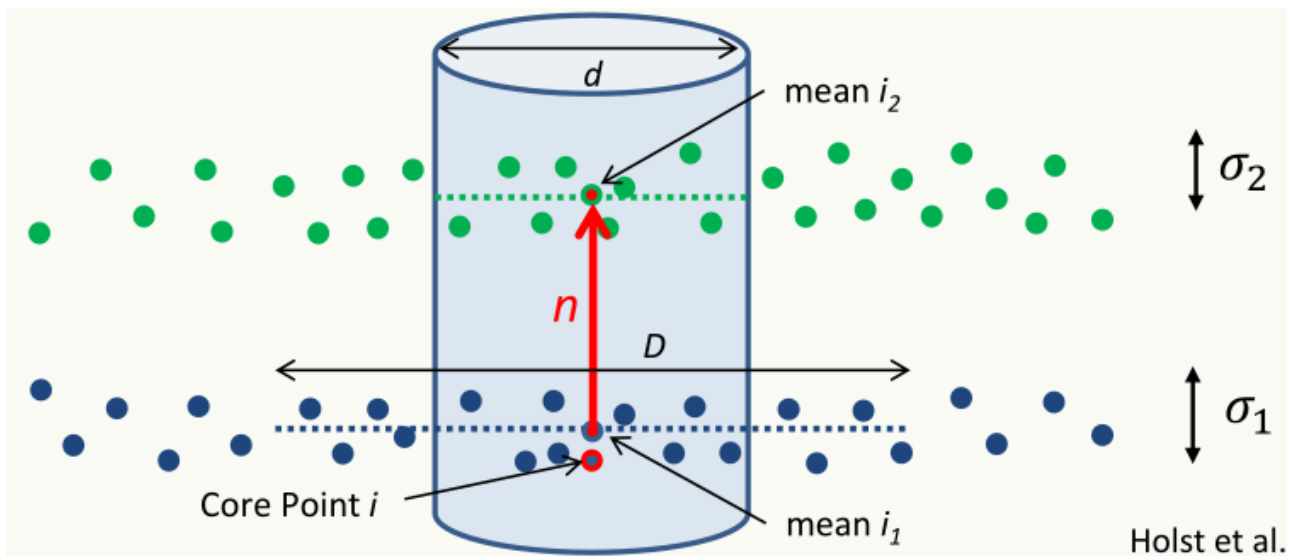
- Detect deformations in

- natural structures (landslides, rockfalls, ...)
- artificial structures (bridges, water dams, buildings)
- Monitor construction sites/mining areas
- Compare constructed objects with plans (e.g. CAD drawing)
- Monitor crop growth

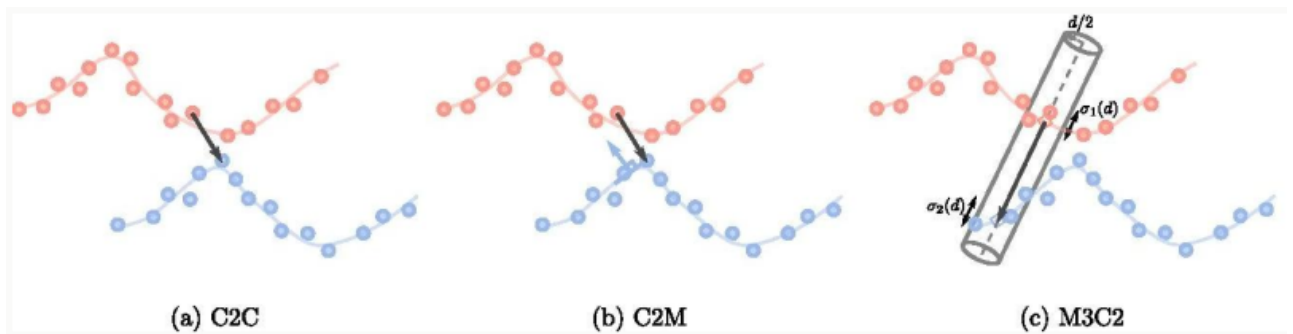


- Cloud to Cloud (C2C)
 - Every point is compared to its nearest neighbour
 - Differences depend on
 - Correspondences (difficult for high curvatures)
 - Point density
 - Outliers
 - Random errors
 - Systematic errors
 - -> **unsigned** distance values
- Cloud to Mesh (C2M)
 - Every point is compared to its nearest plane (perpendicular)
 - Differences depend on
 - Correspondences (slightly improved)
 - Point density
 - Outliers
 - Random errors
 - Systematic errors
 - -> **Signed** distance values
- Multiscale-Model-to-Model-Cloud (M3C2)

- All points within diameter d around a core point define a plane (mean+normal vector)
- Create Cylinder with radius d along the normal vector
- Calculate difference to corresponding mean vector



- Problems: In-Plane Movement



- Correspondences between points change over time
 - All correspondences have a problem
- No differences visible in plane
- Only curved structures show differences